Cross-City Transfer Learning for Data-Scarce Bike-Sharing Demand Forecasting

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1. Problem Statement

Bike-sharing operators require short-term demand forecasts for redistribution and station-capacity planning. New or recently changed systems may, however, have insufficient local history for training reliable deep models. We ask: *Does pretraining on several European bike-sharing systems improve next-hour station-level demand forecasting in a new city when only limited target-city data are available?* Bilbao, Vienna, and Glasgow are used as source systems and Freiburg as the target. We measure transfer gains relative to models trained on the same limited amount of Freiburg data. If transfer fails, the study will still identify negative transfer and estimate how much local history is required before source pretraining stops helping.

2. Existing Methods

Deep bike-sharing forecasting methods combine temporal dependencies with spatial relations. STG2Vec uses heterogeneous spatial-temporal graph representations and recurrent forecasting for New York Citi Bike demand (Li et al., 2019). RegionTrans addresses data scarcity by learning correspondences between data-rich source regions and a target city before transferring spatiotemporal representations (Wang et al., 2019). Our study instead evaluates a simpler, reproducible station-level transfer protocol across four European countries. Because cities have different station sets and network sizes, a shared GRU predicts one scalar demand value per station rather than using a city-specific output vector. Target-history availability is varied explicitly, while unobserved periods are removed through a coverage mask.

3. Data Description

We use the European Bike-Sharing Dataset accompanying Waldner et al. (Waldner et al., 2025). The verified export contains 25,210,627 inferred trips, 267 systems, 13,192 stations, 38,279,885 station-status changes, and 121 bike types. The official repository is public under CC BY-NC 4.0. Trip variables include city, UTC start time, start/end station, coordinates, duration, and distance. Station metadata provide coordinates and rack capacity; city metadata provide timezone and return policy; station status provides bike availability, free racks, and maintenance state.

The common observation interval is 29 August 2022–15 July 2023. The target is the number of valid departures from station s during the next UTC hour. Because status records are change-triggered, missing station-hours are not automatically treated as zero. A zero is retained only when the station-hour is bracketed by status observations within 24 hours, at

Role	System	Station departures	Active stations
Source	Bilbao	1,438,542	44
Source	Vienna	356,447	250
Source	Glasgow	348,612	105
Target	Freiburg	496,145	100

Table 1: Selected systems; screening used only pre-test data.

least 50% of active stations are covered, the city records a status change during the hour, and the station is not under maintenance.

After masking, **2,741,276 station-hours** are eligible for modelling: 816,811 positive-demand hours and 1,924,465 verified zero-demand hours. The final supervised sample count will be slightly lower because every example requires complete lag history within its chronological split.

4. Methodology

For station s and hour t, the shared model predicts

$$\hat{y}_{s,t+1} = f_\theta(\mathbf{y}_{s,t-23:t}, y_{s,t-167}, \mathbf{c}_t, \mathbf{z}_s)\,, \tag{1}$$

where $\mathbf{y}_{s,t-23:t}$ is the previous 24-hour demand sequence, $y_{s,t-167}$ is the same hour one week earlier, $\mathbf{c}_t$ contains calendar and citywide-demand variables, and $\mathbf{z}_s$ contains station attributes.

Each sample uses **34 numerical inputs**: 24 hourly demand values and ten scalar covariates—the previous-week lag, sine/cosine encodings of local hour and weekday, a weekend indicator, normalized latitude and longitude, rack capacity, and recent citywide demand. A GRU is selected because its parameters can be shared across systems with different station counts (Cho et al., 2014).

We compare: (1) historical average, (2) persistence, (3) target-only GRU, (4) pooled GRU trained on source data and available target history, and (5) a source-pretrained GRU fine-tuned on Freiburg. Target adaptation budgets are 1 day, 7 days, 30 days, and the full target-training interval. The chronological split is training (29 Aug 2022–15 Feb 2023), validation (15 Feb–16 Apr 2023), and test (16 Apr–15 Jul 2023). Hyperparameters and early stopping use validation data only.

Implementation will use Python and the open-source PyTorch `torch.nn.GRU` implementation. We will develop the windowing, masking, normalization, pretraining, and fine-tuning pipeline ourselves; DuckDB or Polars will process the Parquet files lazily. Evaluation will report MAE, RMSE, WAPE, errors by station-demand level, and variation across repeated runs. The main result will be a transfer-gain curve over increasing target-history budgets.

References

Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. Learning phrase representations using RNN encoder–decoder for statistical machine translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, pages 1724–1734. Association for Computational Linguistics, 2014. doi: 10.3115/v1/D14-1179.

Youru Li, Zhenfeng Zhu, Deqiang Kong, Meixiang Xu, and Yao Zhao. Learning heterogeneous spatial-temporal representation for bike-sharing demand prediction. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 33, pages 1004–1011, 2019. doi: 10.1609/aaai.v33i01.33011004.

Felix Waldner, Georg Balke, Felix Rech, and Martin Lellep. Data-driven insights into (e-)bike-sharing: mining a large-scale dataset on usage and urban characteristics: descriptive analysis and performance modeling. *Transportation*, 52:2433–2473, 2025. doi: 10.1007/s11116-025-10661-2.

Leye Wang, Xu Geng, Xiaojuan Ma, Feng Liu, and Qiang Yang. Cross-city transfer learning for deep spatio-temporal prediction. In *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence*, pages 1893–1899, 2019. doi: 10.24963/ijcai.2019/262.